ntbk

May 3, 2026

1 Bus Holding Project

1.0.1 Imports + setup

[43]:
```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns
import boto3
from scipy import stats

sns.set_theme(style='whitegrid', context='notebook', palette='deep')
```

1.0.2 Import data

[44]:
```
s3 = boto3.client('s3', region_name='us-east-1')

files = [
    'feeds/nyct_mta_bus_gtfsrt/COMPACTED_nyct_mta_bus_gtfsrt_2023-05-01_13:42:
 ↪41.parquet',
    'feeds/nyct_mta_bus_gtfsrt/COMPACTED_nyct_mta_bus_gtfsrt_2023-05-02_13:42:
 ↪52.parquet',
    'feeds/nyct_mta_bus_gtfsrt/COMPACTED_nyct_mta_bus_gtfsrt_2023-05-03_13:43:
 ↪04.parquet',
    'feeds/nyct_mta_bus_gtfsrt/COMPACTED_nyct_mta_bus_gtfsrt_2023-05-04_13:42:
 ↪58.parquet',
    'feeds/nyct_mta_bus_gtfsrt/COMPACTED_nyct_mta_bus_gtfsrt_2023-05-05_13:42:
 ↪49.parquet',
    'feeds/nyct_mta_bus_gtfsrt/COMPACTED_nyct_mta_bus_gtfsrt_2023-05-08_13:42:
 ↪34.parquet',
    'feeds/nyct_mta_bus_gtfsrt/COMPACTED_nyct_mta_bus_gtfsrt_2023-05-09_13:42:
 ↪56.parquet',
]

import os
os.makedirs('data', exist_ok=True)
```

```
dfs = []
for f in files:
    local = f'data/{f.split("/")[-1]}'
    print(f'Downloading {local}...')
    s3.download_file('busobservatory-lake', f, local)
    dfs.append(pd.read_parquet(local))
    print('Done.')
```

```
Downloading data/COMPACTED_nyct_mta_bus_gtfsrt_2023-05-01_13:42:41.parquet…
Done.
Downloading data/COMPACTED_nyct_mta_bus_gtfsrt_2023-05-02_13:42:52.parquet…
Done.
Downloading data/COMPACTED_nyct_mta_bus_gtfsrt_2023-05-03_13:43:04.parquet…
Done.
Downloading data/COMPACTED_nyct_mta_bus_gtfsrt_2023-05-04_13:42:58.parquet…
Done.
Downloading data/COMPACTED_nyct_mta_bus_gtfsrt_2023-05-05_13:42:49.parquet…
Done.
Downloading data/COMPACTED_nyct_mta_bus_gtfsrt_2023-05-08_13:42:34.parquet…
Done.
Downloading data/COMPACTED_nyct_mta_bus_gtfsrt_2023-05-09_13:42:56.parquet…
Done.
```

[45]:
```
df = pd.concat(dfs, ignore_index=True)
df.head()
```

[45]:
```
                id          vehicle.trip.trip_id vehicle.trip.start_date  …
vehicle.timestamp  vehicle.stop_id  vehicle.vehicle.id
0  MTA NYCT_4093    FP_B3-Sunday-054100_B7_602                20230430  …
2023-04-30 13:40:37+00:00           307983       MTA NYCT_4093
1  MTA NYCT_5331   OH_B3-Sunday-057500_M101_42                20230430  …
2023-04-30 13:40:18+00:00           404338       MTA NYCT_5331
2  MTA NYCT_5330   OH_B3-Sunday-052500_M15_217                20230430  …
2023-04-30 13:40:28+00:00           404844       MTA NYCT_5330
3  MTA NYCT_7989    UP_B3-Sunday-058300_B3_109                20230430  …
2023-04-30 13:40:41+00:00           305480       MTA NYCT_7989
4  MTA NYCT_5325  WF_B3-Sunday-056000_BX35_870                20230430  …
2023-04-30 13:40:27+00:00           102328       MTA NYCT_5325

[5 rows x 11 columns]
```

[46]:
```
df['vehicle.trip.route_id'].value_counts().head(20)
```

[46]:
```
vehicle.trip.route_id
B6      271400
Q58     227172
B41     214530
Q44+    195138
```

```
Q27      190458
M101     177166
M4       159869
M15+     157186
BX36     153904
B35      149047
B15      143996
Q46      142282
BX9      140037
B38      139084
BX19     137970
Q25      137797
Q65      136750
B46      135671
M3       133269
B103     133094
Name: count, dtype: int64
```

1.0.3 Clean and filter data

[47]:
```
m15 = df[df['vehicle.trip.route_id'] == 'M15'].copy() # bus route we're␣
↪modelling
m15 = m15[m15['vehicle.trip.direction_id'] == 1.0].copy() # make sure they're␣
↪all going in the same direction (higher traffic)
m15['time'] = pd.to_datetime(m15['vehicle.timestamp'], unit='s')

m15
```

[47]:
```
                      id                vehicle.trip.trip_id
vehicle.trip.start_date  … vehicle.stop_id  vehicle.vehicle.id
time
2          MTA NYCT_5330        OH_B3-Sunday-052500_M15_217
20230430  …           404844       MTA NYCT_5330 2023-04-30 13:40:28+00:00
642        MTA NYCT_5921        OH_B3-Sunday-051200_M15_218
20230430  …           405359       MTA NYCT_5921 2023-04-30 13:40:31+00:00
645        MTA NYCT_5915        OH_B3-Sunday-056100_M15_219
20230430  …           401756       MTA NYCT_5915 2023-04-30 13:40:26+00:00
646        MTA NYCT_5919        OH_B3-Sunday-057300_M15_205
20230430  …           803228       MTA NYCT_5919 2023-04-30 13:40:32+00:00
911        MTA NYCT_6067        OH_B3-Sunday-054900_M15_212
20230430  …           401765       MTA NYCT_6067 2023-04-30 13:40:27+00:00
…                   …                                 …
…  …               …                  …                          …
18636881  MTA NYCT_5891  OH_B3-Weekday-SDon-052300_M15_229
20230509  …           401748       MTA NYCT_5891 2023-05-09 13:03:17+00:00
18636884  MTA NYCT_5896  OH_B3-Weekday-SDon-045900_M15_210
20230509  …           404844       MTA NYCT_5896 2023-05-09 13:03:22+00:00
```

```
18637819  MTA NYCT_5927  OH_B3-Weekday-SDon-053900_M15_212
20230509  …          401742       MTA NYCT_5927 2023-05-09 13:03:27+00:00
18637823  MTA NYCT_5910  OH_B3-Weekday-SDon-046700_M15_223
20230509  …          403849       MTA NYCT_5910 2023-05-09 13:03:29+00:00
18638459  MTA NYCT_6055  OH_B3-Weekday-SDon-053100_M15_209
20230509  …          401748       MTA NYCT_6055 2023-05-09 13:03:30+00:00

[70862 rows x 12 columns]
```

[48]:
```python
# Filter to peak weekday hours
m15['hour'] = m15['time'].dt.hour
m15['weekday'] = m15['time'].dt.weekday
m15 = m15[
    (m15['weekday'] < 5) &
    ((m15['hour'].between(7, 9)) | (m15['hour'].between(16, 19)))
].copy()

m15
```

[48]:
```
                      id                vehicle.trip.trip_id
vehicle.trip.start_date vehicle.trip.route_id  …  vehicle.vehicle.id
time  hour  weekday
1309225   MTA NYCT_5870  OH_B3-Weekday-SDon-015000_M15_202
20230501                   M15  …       MTA NYCT_5870 2023-05-01
07:00:50+00:00     7        0
1309433   MTA NYCT_5870  OH_B3-Weekday-SDon-015000_M15_202
20230501                   M15  …       MTA NYCT_5870 2023-05-01
07:01:21+00:00     7        0
1309643   MTA NYCT_5870  OH_B3-Weekday-SDon-015000_M15_202
20230501                   M15  …       MTA NYCT_5870 2023-05-01
07:02:21+00:00     7        0
1309859   MTA NYCT_5870  OH_B3-Weekday-SDon-015000_M15_202
20230501                   M15  …       MTA NYCT_5870 2023-05-01
07:03:22+00:00     7        0
1310074   MTA NYCT_5870  OH_B3-Weekday-SDon-015000_M15_202
20230501                   M15  …       MTA NYCT_5870 2023-05-01
07:04:23+00:00     7        0
…                 …                                  …
…                   …  …                  …                          …
…      …
17954841  MTA NYCT_5927  OH_B3-Weekday-SDon-035800_M15_212
20230509                   M15  …       MTA NYCT_5927 2023-05-09
09:59:30+00:00     9        1
18259589  MTA NYCT_5867  OH_B3-Weekday-SDon-023000_M15_205
20230509                   M15  …       MTA NYCT_5867 2023-05-09
07:53:35+00:00     7        1
18259653  MTA NYCT_5926  OH_B3-Weekday-SDon-021000_M15_204
```

```
20230509                    M15  …       MTA NYCT_5926 2023-05-09
07:53:15+00:00     7         1
18259818  MTA NYCT_5867  OH_B3-Weekday-SDon-023000_M15_205
20230509                    M15  …       MTA NYCT_5867 2023-05-09
07:54:37+00:00     7         1
18259885  MTA NYCT_5926  OH_B3-Weekday-SDon-021000_M15_204
20230509                    M15  …       MTA NYCT_5926 2023-05-09
07:54:46+00:00     7         1

[15652 rows x 14 columns]
```

Compute the distance gaps between consecutive buses. At each 1-minute snapshot we sort active buses by latitude and emit one row per consecutive `(leader, follower)` pair, tagged by both vehicle IDs. This lets us reconstruct per-pair gap trajectories — actual sample paths of the OU process from §2.1 — rather than a fleet-level spatial average. Each observation is now the kind of quantity a dispatcher would observe and act on, and matches the state variable of the controller in §3.

[49]:
```
m15 = m15.sort_values(['time', 'vehicle.position.latitude'], ascending=[True,␣
↪False])
m15['minute'] = m15['time'].dt.floor('1min')  # 1-min time buckets for grouping␣
↪snapshots

# At each 1-min snapshot, sort active M15 northbound buses N->S and emit one␣
↪row per
# consecutive (leader, follower) pair, tagged by both vehicle IDs. This␣
↪preserves
# pair identity so we can later string together each pair's gap trajectory.
records = []
for minute, group in m15.groupby('minute'):
    g = (group.drop_duplicates('vehicle.vehicle.id')
              .sort_values('vehicle.position.latitude', ascending=False))
    if len(g) < 2:
        continue
    vids = g['vehicle.vehicle.id'].values
    lats = g['vehicle.position.latitude'].values
    for i in range(len(vids) - 1):
        records.append({
            'minute':   minute,
            'leader':   vids[i],          # bus to the north
            'follower': vids[i + 1],      # bus to the south
            'gap_km':   (lats[i] - lats[i + 1]) * 111  # lat-degree -> km
        })

pairs_df = pd.DataFrame(records)
pairs_df.head(10)
```

[49]:

	minute	leader	follower	gap_km
0	2023-05-01 07:06:00+00:00	MTA NYCT_5834	MTA NYCT_5870	9.079216
1	2023-05-01 07:07:00+00:00	MTA NYCT_5834	MTA NYCT_5870	8.506737
2	2023-05-01 07:08:00+00:00	MTA NYCT_5834	MTA NYCT_5870	8.032917
3	2023-05-01 07:09:00+00:00	MTA NYCT_5834	MTA NYCT_5870	8.006241
4	2023-05-01 07:10:00+00:00	MTA NYCT_5834	MTA NYCT_5870	7.961781
5	2023-05-01 07:11:00+00:00	MTA NYCT_5834	MTA NYCT_5870	7.681892
6	2023-05-01 07:12:00+00:00	MTA NYCT_5834	MTA NYCT_5870	7.280903
7	2023-05-01 07:13:00+00:00	MTA NYCT_5834	MTA NYCT_5870	7.243641
8	2023-05-01 07:14:00+00:00	MTA NYCT_5834	MTA NYCT_5870	7.108566
9	2023-05-01 07:15:00+00:00	MTA NYCT_5834	MTA NYCT_5870	6.664810

Build (G_t, G_{t+1}) regression pairs from per-pair trajectories. We only keep observations where the *same* (leader, follower) pair was consecutive at both minute t and minute $t+1$ — so each regression pair represents one bus following another bus from one minute to the next, never spanning a "pair break" (where a third bus enters between them, or one bus ends its trip). This pools across all pair-trajectories present in the filtered dataset, treating each pair as an i.i.d. sample of the same OU process under the stationarity-during-peak-hours assumption.

[50]:

```
# Sort by pair, then by minute. Within each (leader, follower) group, shift the
# next minute and next gap forward; a regression pair (G_t, G_{t+1}) is valid
# iff the next observation in the same pair group is exactly 1 minute later.
pairs_df = pairs_df.sort_values(['leader', 'follower', 'minute']).
 ↪reset_index(drop=True)
pairs_df['next_minute'] = pairs_df.groupby(['leader', 'follower'])['minute'].
 ↪shift(-1)
pairs_df['next_gap']    = pairs_df.groupby(['leader', 'follower'])['gap_km'].
 ↪shift(-1)

# Adjacent-minute pairs only; drop physically implausible gaps (overlap /␣
 ↪route-edge artifacts)
mask  = (pairs_df['next_minute'] - pairs_df['minute']) == pd.Timedelta('1min')
mask &= pairs_df['gap_km'].between(0.05, 4.0)
mask &= pairs_df['next_gap'].between(0.05, 4.0)

ar1_pairs = pairs_df.loc[mask, ['minute', 'leader', 'follower', 'gap_km',␣
 ↪'next_gap']]

print(f"Total pair-minute rows:                 {len(pairs_df)}")
print(f"Distinct (leader, follower) pairs:      {pairs_df.
 ↪groupby(['leader','follower']).ngroups}")
print(f"AR(1) regression pairs (G_t, G_{{t+1}}):  {len(ar1_pairs)}")
print()
print(ar1_pairs['gap_km'].describe())
```

```
Total pair-minute rows:                 13062
Distinct (leader, follower) pairs:      446
AR(1) regression pairs (G_t, G_{t+1}):  10076
```

```
count    10076.000000
mean         1.361185
std          0.886808
min          0.050388
25%          0.664258
50%          1.205509
75%          1.952442
max          3.993805
Name: gap_km, dtype: float64
```

Fit AR(1) regression to confirm autocorrelation

[51]:
```
# Fit AR(1) on the pooled (G_t, G_{t+1}) pairs from per-pair trajectories.
G_t  = ar1_pairs['gap_km'].values
G_t1 = ar1_pairs['next_gap'].values

slope, intercept, r_value, p_value, std_err = stats.linregress(G_t, G_t1)
print(f"AR(1) fit (per-pair, 1-min sampling):")
print(f"  beta  = {slope:.4f}")
print(f"  R²    = {r_value**2:.4f}")
print(f"  p     = {p_value:.4e}")
print(f"  n     = {len(G_t)}")
```

```
AR(1) fit (per-pair, 1-min sampling):
  beta  = 0.9820
  R²    = 0.9789
  p     = 0.0000e+00
  n     = 10076
```

A high β (close to 1) indicates strong autocorrelation — each per-pair gap behaves as a slowly mean-reverting process, consistent with the OU model. The R^2 tells us how much variance the AR(1) discretization captures; the residual variance is what feeds the diffusion coefficient σ.

1.0.4 Plot and Verify Mean-Reversion Assumption

[52]:
```
residuals = G_t1 - (intercept + slope * G_t)
s = residuals.std()

dt = 1.0  # 1-minute sampling interval (per-pair, adjacent minutes)
kappa = -np.log(max(slope, 1e-10)) / dt
G_bar = intercept / (1 - slope)
sigma = s * np.sqrt(2 * kappa / (1 - slope**2))

print(f"OU Parameters (per-pair, 1-min):")
print(f"  kappa = {kappa:.4f} per minute")
print(f"  G_bar = {G_bar:.4f} km")
print(f"  sigma = {sigma:.4f} km/√min")
```

```
OU Parameters (per-pair, 1-min):
  kappa = 0.0182 per minute
  G_bar = 0.9414 km
  sigma = 0.1289 km/√min
```

Fitted Params κ: mean-reversion speed (per minute). The half-life of a gap deviation is $\ln(2)/\kappa$ minutes — how fast a perturbed gap returns to equilibrium.

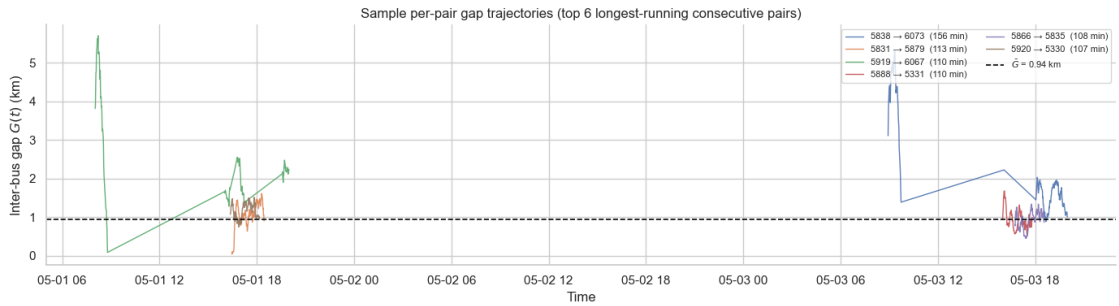
$\bar{G}$ (km): the equilibrium per-pair spacing. Inter-bus gaps between a fixed leader–follower pair are pulled toward this during peak hours.

σ (km/√min): the per-pair noise scale. Traffic, signal timing, boarding randomness — everything we can't predict that pushes the leader–follower gap around between minute ticks.

[53]:
```
# Plot the longest-running per-pair trajectories. Each line is one (leader,␣
↪follower)
# pair's gap G(t) in km, sampled every minute they remained consecutive.
pair_lengths = (pairs_df.groupby(['leader', 'follower']).size()
                        .sort_values(ascending=False))
top_pairs = pair_lengths.head(6).index.tolist()

fig, ax = plt.subplots(figsize=(14, 4))
palette = sns.color_palette('deep', n_colors=len(top_pairs))
for (leader, follower), color in zip(top_pairs, palette):
    traj = (pairs_df[(pairs_df['leader'] == leader) &
                     (pairs_df['follower'] == follower)]
            .set_index('minute')['gap_km'].sort_index())
    ax.plot(traj.index, traj.values, linewidth=1.0, color=color, alpha=0.9,
            label=f'{leader[-4:]} → {follower[-4:]}  ({len(traj)} min)')

ax.axhline(G_bar, linestyle='--', color='black', linewidth=1.2,
           label=f'$\\bar{{G}}$ = {G_bar:.2f} km')
ax.set_xlabel('Time')
ax.set_ylabel('Inter-bus gap $G(t)$ (km)')
ax.set_title('Sample per-pair gap trajectories (top 6 longest-running␣
↪consecutive pairs)')
ax.legend(loc='upper right', fontsize=8, ncol=2)
sns.despine()
plt.tight_layout()
plt.show()
```



1.0.5 HJB Solver

[54]:
```
dt_solver = 0.005  # small enough for advection + diffusion stability

# Cost weights
alpha = 1.0 # bunching penalty
beta = 0.01 # holding penalty
tau_max = 0.5 # max holding rate (km/min)

dG = 0.1 # spatial step

# Grid
G_grid = np.arange(0.0, 6.0 + dG, dG)
N_G = len(G_grid)

# Time Horizon
T = 120 # two hour peak period
t_grid = np.arange(0, T + dt_solver, dt_solver)
N_t = len(t_grid)
```

Check the CFL (Courant-Friedrichs-Lewy) to verify stability condition. For advection-diffusion PDEs, both terms must be stable:

Diffusion: $\frac{\sigma^2 \Delta t}{\Delta G^2} \leq \frac{1}{2}$

Advection: $\frac{|\mu_{\max}| \Delta t}{\Delta G} \leq 1$

where $\mu_{\max} = \kappa \cdot G_{\max} + \tau_{\max}$ is the worst-case drift.

[55]:
```
cfl_diff = sigma**2 * dt_solver / dG**2
mu_max = kappa * G_grid.max() + tau_max
cfl_adv = mu_max * dt_solver / dG

print(f"Diffusion CFL:  {cfl_diff:.4f} - {'STABLE' if cfl_diff <= 0.5 else␣
↪'UNSTABLE'}")
print(f"Advection CFL:  {cfl_adv:.4f} - {'STABLE' if cfl_adv <= 1.0 else␣
↪'UNSTABLE'}")
```

```
Diffusion CFL:  0.0083 - STABLE
Advection CFL:  0.0305 - STABLE
```

Initialize value function Terminal condition: $V(G,T) = 0 \forall G$ - no cost remaining at end of horizon

$V(G,t)$ represents the minimum expected cost from now until T given you're currently at gap G at time t.

[56]:
```
V = np.zeros((N_G, N_t))
tau_star = np.zeros((N_G, N_t))
```

Backward Sweep Solve HJB backwards from $t = T$ to $t = 0$ At each step: approximate spatial derivatives via finite differences, solve bang-bang condition for τ^*, update V^k explicitly

[57]:
```
# Solver

for k in range(N_t - 2, -1, -1):
    V_k1 = V[:, k + 1]

    # First derivative (central differences interior, 3-point at boundaries)
    dV = np.zeros(N_G)
    dV[1:-1] = (V_k1[2:] - V_k1[:-2]) / (2 * dG)
    dV[0] = (-3 * V_k1[0] + 4 * V_k1[1] - V_k1[2]) / (2 * dG)
    dV[-1] = (3 * V_k1[-1] - 4 * V_k1[-2] + V_k1[-3]) / (2 * dG)

    # Second derivative (midpoint formula)
    d2V = np.zeros(N_G)
    d2V[1:-1] = (V_k1[2:] - 2 * V_k1[1:-1] + V_k1[:-2]) / dG**2
    d2V[0] = d2V[1]
    d2V[-1] = d2V[-2]

    # Bang-bang optimal policy: minimize tau*(beta - dV)
    # hold (tau_max) when dV > beta, i.e. cost increases with gap
    tau = np.where(dV >= beta, tau_max, 0.0)
    tau_star[:, k] = tau

    # Explicit backward update
    V[:, k] = V_k1 + dt_solver * (
        alpha * (G_bar - G_grid)**2
        + beta * tau + (kappa *
        (G_bar - G_grid) - tau)
        * dV + 0.5 * sigma**2 * d2V
    )

print("Solver complete.")
print(f"V range:    [{V.min():.2f}, {V.max():.2f}]")
print(f"tau* range: [{tau_star.min():.2f}, {tau_star.max():.2f}]")
```

```
Solver complete.
V range:    [0.00, 99.42]
tau* range: [0.00, 0.50]
```

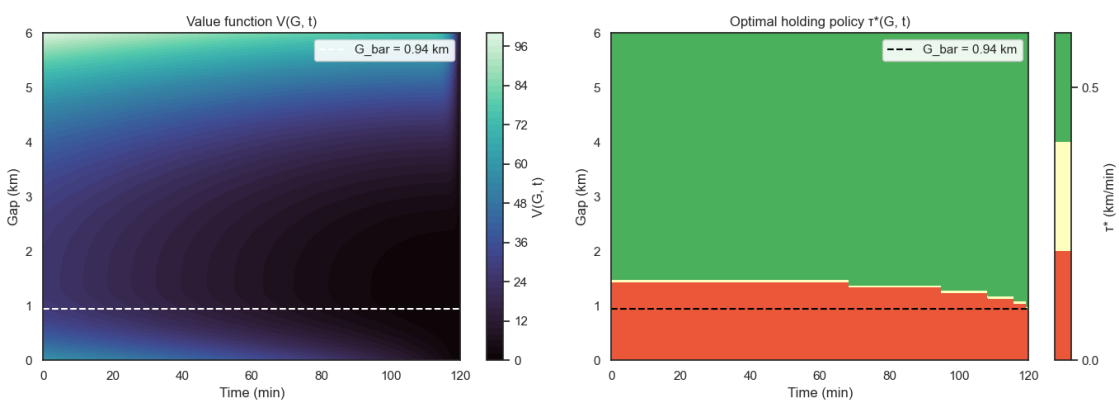
Plot value function and optimal policy

[58]:
```
with sns.axes_style('white'):
    fig, axes = plt.subplots(1, 2, figsize=(14, 5))

    im1 = axes[0].contourf(t_grid, G_grid, V, levels=50, cmap='mako')
    cbar1 = plt.colorbar(im1, ax=axes[0])
    cbar1.set_label('V(G, t)')
    axes[0].axhline(G_bar, color='white', linestyle='--', linewidth=1.5,
                    label=f'G_bar = {G_bar:.2f} km')
    axes[0].set_xlabel('Time (min)')
    axes[0].set_ylabel('Gap (km)')
    axes[0].set_title('Value function V(G, t)')
    axes[0].legend(loc='upper right', framealpha=0.9)

    im2 = axes[1].contourf(t_grid, G_grid, tau_star, levels=2, cmap='RdYlGn')
    cbar2 = plt.colorbar(im2, ax=axes[1], ticks=[0, tau_max])
    cbar2.set_label(' * (km/min)')
    axes[1].axhline(G_bar, color='black', linestyle='--', linewidth=1.5,
                    label=f'G_bar = {G_bar:.2f} km')
    axes[1].set_xlabel('Time (min)')
    axes[1].set_ylabel('Gap (km)')
    axes[1].set_title('Optimal holding policy  *(G, t)')
    axes[1].legend(loc='upper right', framealpha=0.9)

    plt.tight_layout()
    plt.show()
```



1.0.6 Monte Carlo Simulation

Simulate 1000 OU paths forward in time. At each step, we apply the optimal policy by looking up $\tau^*(G, t)$ from the precomputed grid. Compare total cost against a no-holding baseline.

[59]:
```
np.random.seed(42)
N_paths = 1000
G0 = G_bar  # start at equilibrium

costs_opt      = np.zeros(N_paths)
costs_baseline = np.zeros(N_paths)

for n in range(N_paths):
    G_opt  = G0
    G_base = G0
    cost_opt  = 0.0
    cost_base = 0.0

    for k in range(N_t - 1):
        Z = np.random.randn()

        # Look up optimal tau from precomputed grid
        i = np.argmin(np.abs(G_grid - G_opt))
        tau = tau_star[i, k]

        # Accumulate costs
        cost_opt  += dt_solver * (alpha * (G_bar - G_opt)**2  + beta * tau)
        cost_base += dt_solver * (alpha * (G_bar - G_base)**2)

        # Evolve both paths with same random shock
        G_opt  = G_opt  + (kappa*(G_bar - G_opt)  - tau)*dt_solver + sigma*np.
 ↪sqrt(dt_solver)*Z
        G_base = G_base +  kappa*(G_bar - G_base)       *dt_solver + sigma*np.
 ↪sqrt(dt_solver)*Z

        # Keep in bounds
        G_opt  = np.clip(G_opt,  G_grid[0], G_grid[-1])
        G_base = np.clip(G_base, G_grid[0], G_grid[-1])

    costs_opt[n]      = cost_opt
    costs_baseline[n] = cost_base

print(f"Mean cost - optimal policy:  {costs_opt.mean():.4f}")
print(f"Mean cost - no holding:      {costs_baseline.mean():.4f}")
print(f"Cost reduction:              {(1 - costs_opt.mean()/costs_baseline.
 ↪mean())*100:.1f}%")
```

```
Mean cost - optimal policy:  17.4817
Mean cost - no holding:      32.1523
Cost reduction:              45.6%
```

1.0.7 Cost Distribution

[60]:
```
fig, axes = plt.subplots(1, 2, figsize=(14, 4))

palette = sns.color_palette('deep')
baseline_color = palette[3]   # red
optimal_color = palette[2]    # green

# Cost distribution - KDE-overlaid histograms
sns.histplot(costs_baseline, bins=50, ax=axes[0], color=baseline_color,
             alpha=0.5, kde=True, label='No holding', stat='count',
             edgecolor='none')
sns.histplot(costs_opt, bins=50, ax=axes[0], color=optimal_color,
             alpha=0.5, kde=True, label='Optimal policy', stat='count',
             edgecolor='none')
axes[0].axvline(costs_baseline.mean(), color=baseline_color, linestyle='--',␣
 ↪linewidth=1.5)
axes[0].axvline(costs_opt.mean(), color=optimal_color, linestyle='--',␣
 ↪linewidth=1.5)
axes[0].set_xlabel('Total cost')
axes[0].set_ylabel('Count')
axes[0].set_title('Cost distribution: optimal vs. baseline')
axes[0].legend()

# Sample paths
axes[1].set_title('Sample gap paths: optimal vs. baseline')
opt_label_used = base_label_used = False
for n in range(5):
    np.random.seed(n)
    G_opt  = G0
    G_base = G0
    path_opt  = [G_opt]
    path_base = [G_base]
    for k in range(N_t - 1):
        Z = np.random.randn()
        i = np.argmin(np.abs(G_grid - G_opt))
        tau = tau_star[i, k]
        G_opt  = np.clip(G_opt  + (kappa*(G_bar-G_opt) -tau)*dt_solver +␣
 ↪sigma*np.sqrt(dt_solver)*Z, G_grid[0], G_grid[-1])
        G_base = np.clip(G_base +  kappa*(G_bar-G_base)    *dt_solver +␣
 ↪sigma*np.sqrt(dt_solver)*Z, G_grid[0], G_grid[-1])
        path_opt.append(G_opt)
        path_base.append(G_base)
```

```
    sns.lineplot(x=t_grid, y=path_opt, ax=axes[1], color=optimal_color,
                 alpha=0.6, linewidth=1.0,
                 label='Optimal policy' if not opt_label_used else None)
    sns.lineplot(x=t_grid, y=path_base, ax=axes[1], color=baseline_color,
                 alpha=0.6, linewidth=1.0,
                 label='No holding' if not base_label_used else None)
    opt_label_used = base_label_used = True

axes[1].axhline(G_bar, color='black', linestyle='--', linewidth=1.5,
                label=f'G_bar = {G_bar:.2f}')
axes[1].set_xlabel('Time (min)')
axes[1].set_ylabel('Gap (km)')
axes[1].legend()

sns.despine(fig=fig)
plt.tight_layout()
plt.show()
```

```
/opt/anaconda3/lib/python3.11/site-packages/seaborn/_oldcore.py:1119:
FutureWarning: use_inf_as_na option is deprecated and will be removed in a
future version. Convert inf values to NaN before operating instead.
  with pd.option_context('mode.use_inf_as_na', True):
/opt/anaconda3/lib/python3.11/site-packages/seaborn/_oldcore.py:1119:
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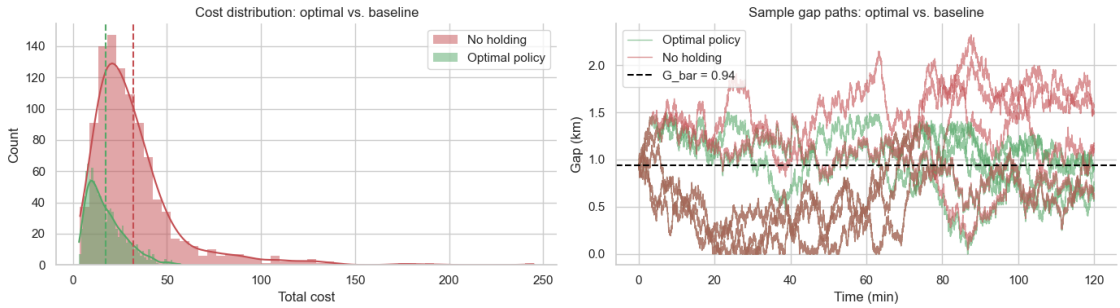
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